

BREAK INTO AEROSPACE

Structural Design Engineer

Subdomain Interview Preparation Guide

Targeted for the 2025–2026 Recruitment Cycle

What This Document Covers: Aerospace structural design engineering is simultaneously the most visually intuitive and the most analytically demanding discipline in the aircraft or spacecraft development process. At its broadest, the term encompasses the full range of activities through which an engineer translates a structural requirement — a load case, a fatigue life, a weight budget, a stiffness target — into a physical geometry: the shape of a rib, the thickness of a skin panel, the routing of a spar, the arrangement of fasteners that join them, and the CAD model that captures all of it with sufficient precision for manufacturing, analysis, and certification. At its most specific, it is the discipline of making the geometry work — ensuring that a structural member fits within the envelope defined by aerodynamics and systems, that the joints are accessible for manufacture and inspection, that the load paths are continuous through the structure, and that every feature of the physical design can be substantiated by stress analysis, tested by coupon and component specimens, and certified by the authority with the appropriate means of compliance.

The engineering challenges that define this sector are genuinely multi-domain. The structure must be strong enough to carry the design limit load without yielding and the design ultimate load without failing, but light enough to satisfy the weight budget that aircraft economics require — a tension that is resolved not by optimising either constraint alone, but by finding the design that simultaneously satisfies both with the minimum possible material. The geometry must be manufacturable — which means understanding the capabilities and limitations of the specific manufacturing process (whether machining, sheet metal forming, composite layup, additive manufacture, or casting) and designing features that can be produced with the required tolerances. The structure must be inspectable throughout its service life — which means designing for access, designing for the non-destructive inspection technique that will be used, and understanding what the inspection capability determines about the damage tolerance life of the component. And the entire design must be captured in a CAD model that is the single source of truth for manufacturing, analysis, assembly, and certification — a model that must be created, managed, and updated with a discipline that has more in common with software configuration management than with traditional drawing practice.

The employer landscape spans the major airframe original equipment OEMs who employ the largest structural design engineering populations (Boeing — Commercial Airplanes and Defense, Space & Security; Airbus — Commercial Aircraft, Defence and Space, and Helicopters; Lockheed Martin Aeronautics; Northrop Grumman Aeronautics Systems; Raytheon Technologies; Leonardo Aircraft; Embraer; Bombardier; Textron Aviation), the aerostructures suppliers who produce major structural components to OEM design or their own design under risk-sharing partnerships (Spirit AeroSystems, GKN Aerospace, Safran Nacelles, Collins Aerospace Aerostructures, Ducommun, Triumph Group, Senior Aerospace), the spacecraft and launch vehicle developers where structural design intersects with thermal, propulsion, and mechanisms engineering (SpaceX Structures, Blue Origin, Rocket Lab, Airbus Defence and Space, Thales Alenia Space, NASA in-house centres including Langley, Marshall, and Johnson), the defence prime contractors where structural design operates under significantly more stringent security and classification environments (BAE Systems, General Dynamics, MBDA, Dassault Aviation), the engineering services and design consultancies that provide structural design capacity to OEMs and aerostructures suppliers (Expleo, Akka Technologies — now Akkodis, Alten, Assystem, HCL Aviotech), and the increasingly active advanced air mobility developers whose novel vehicle configurations demand structural design engineering for configurations that have no established design handbook solutions (Joby Aviation, Archer Aviation, Wisk Aero, Beta Technologies).

The technical baseline is shared across all of these employers — the mechanics of deformable solids, the fatigue behaviour of metallic and composite materials, the geometric constraints of assembly — but the specific design challenges, the CAD environment, the certification context, and the career trajectory differ substantially between a junior structural designer at an aerostructures supplier and a senior structural design engineer leading the wing box design at an OEM.

PART I: WHAT STRUCTURAL DESIGN ENGINEERING ACTUALLY LOOKS FOR

By Level

Intern / Graduate

At entry level, structural design engineering employers are not primarily screening for advanced structural theory — they are screening for strong spatial reasoning and three-dimensional geometric thinking, CAD proficiency in the relevant platform (CATIA V5/V6, Siemens NX, or SolidWorks depending on the employer), and the specific intellectual quality that structural design most values: the ability to visualise a three-dimensional structure from a two-dimensional drawing, understand how it assembles, identify where the interfaces are, and anticipate where the design will be difficult to manufacture, inspect, or modify.

This geometric intuition is specifically important because structural design engineering is not primarily a computational discipline at the junior level — it is a spatial discipline. The stress analyst computes loads and margins; the structural designer translates those computed requirements into geometry that can be built and assembled. The designer who cannot visualise how the rib fits between the forward and rear spars, how the skin panel is fastened to the rib flange, and whether the fastener pattern leaves enough edge distance for the calculated bearing load will produce a design that the stress analyst approves on paper but that the manufacturing team cannot build, or that is structurally sound but unmaintainable in service. The ability to think in three dimensions — to see a complex assembly as a collection of individually simple parts that fit together in a specific sequence — is the spatial intelligence that structural design employers specifically screen for.

Graduate interviews at OEMs and aerostructures suppliers typically involve a combination of CAD competency assessment (a modelling exercise in the employer's preferred platform, or a discussion of the candidate's previous modelling experience and the complexity of the models they have created), geometric problem-solving (spatial reasoning questions, tolerance stackup scenarios, or assembly sequencing problems), and structural fundamentals (beam bending, shear flow in thin-walled sections, the difference between buckling and material failure). At defence primes and spacecraft developers, the screen tends to place additional weight on understanding of drawing standards (AS9100, GD&T per ASME Y14.5), configuration management discipline, and the candidate's experience with the specific CAD platform used in the employer's design environment.

Mid-Level (Design Engineer, Senior Design Engineer, ~3–8 years)

At mid-level, structural design engineering interviews become component-specific and integration-focused. The sector's defining challenge at mid-level is interface management: not the design of any individual structural component in isolation, but the management of the geometric and load interfaces between it and every adjacent component in the assembly. The structural design engineer at this level is the person who owns the geometry at the interfaces — who ensures that the forward spar of the wing box meets the fuselage frame at the correct location, angle, and load-transfer mechanism, and that the resulting joint is simultaneously structurally efficient, manufacturable to tolerance, and capable of being disassembled for inspection or repair.

The specific question that most reliably reveals mid-level structural design depth is about the design conflict: “your rib design calls for a flange thickness of 4mm at the lower skin interface, but the stress analysis shows you need 6mm to meet the ultimate load requirement without yielding, the weight budget allows you 4.5mm maximum, and the tooling for the current geometry is already ordered — walk me through how you think about resolving this conflict, what options you would explore, and what the process is for agreeing a change with the stress team, the manufacturing team, and the weight engineer.” The candidate who can reason through the material grade change, the local pad-up option, the fastener pattern optimisation, the tooling impact of each option, and the change management process that governs a geometry modification after tooling release is demonstrating the design integration maturity that hiring managers at OEMs and aerostructures suppliers specifically assess.

Senior (Principal Engineer, Lead Designer, Chief of Structure)

At senior level, structural design engineering interviews assess the ability to lead the architecture of a major structural assembly — to define the load path philosophy, to establish the interface control documents, to manage the design across a team of junior and mid-level designers, and to represent the structural design position in the integration reviews where aerodynamics, systems, manufacturing, and weight all have competing claims on the same geometric envelope.

The distinguishing credential at senior level is having led the structural design of a major assembly — a wing box, a fuselage section, a primary flight structure — from initial architecture through detailed design to first

article inspection and, ideally, through the certification substantiation that closes the design. Engineers who have done this, and who can describe the specific architectural decisions they made — the load path choices, the interface definitions, the material selections, the joint philosophies — and how those decisions were tested by the integration process and the certification review, have the design leadership experience that is genuinely rare and increasingly valuable as the commercial aviation production rate recovery and the defence modernisation programmes drive demand for experienced structural design talent.

Hands-On Experience and Domain Knowledge Treated as Strong Signals

- **CATIA V5/V6 and the model-based definition environment — the CAD platform that dominates aerospace structural design.** CATIA (Computer Aided Three-dimensional Interactive Application) — developed by Dassault Systèmes and used as the primary structural design platform by Airbus, Boeing (on the 787 and subsequent programmes), Spirit AeroSystems, GKN Aerospace, and the majority of the tier-one aerostructures supply chain — is the de facto standard for aerospace structural design. Understanding CATIA at the level of its specific structural design workflows: Part Design for solid modelling of machined components, Generative Shape Design for surface-driven structural geometry, Assembly Design for managing the parametric relationships between components in an assembly, and Drafting for the production of engineering drawings to AS9100 standards. The shift to model-based definition (MBD) — in which the 3D CAD model carries the full engineering definition (tolerances, materials, surface finish, inspection requirements) rather than the 2D drawing — is the workflow transformation that most significantly affects how structural design engineering is practised today, and engineers who have worked in a genuine MBD environment understand the configuration management discipline that the model-as-master approach requires.
- **Geometric dimensioning and tolerancing — the language through which design intent is communicated to manufacturing and inspection.** GD&T (Geometric Dimensioning and Tolerancing), standardised under ASME Y14.5, is the formal language through which structural designers communicate the permissible variation in a component's geometry to the manufacturing and inspection teams. Understanding GDT at the working level — what a flatness tolerance means and how it differs from a parallelism tolerance, how a true position callout for a fastener hole pattern is interpreted, what a datum reference frame is and how it is established, and how tolerances stack up through an assembly to produce a gap or interference at an interface — is the design communication competency that manufacturing engineers, stress analysts, and tooling designers rely on when they receive a structural design package. Engineers who have applied GD&T correctly on real components — and who have navigated the conversation with the manufacturing team about which tolerances are achievable with the selected process and which require an engineering concession or a process change — are demonstrating the practical manufacturing interface competency that structural design roles require.
- **Thin-walled aerospace structures — spars, ribs, frames, skin panels, and the load path through them.** The primary structural elements of an aircraft airframe — the wing box (spars, ribs, stringers, and skin panels), the fuselage (frames, longerons, stringers, and skin), the empennage, and the primary attachment fittings — are thin-walled structures in which the structural efficiency comes from the arrangement of material at the outer surface of the cross-section rather than its concentration at the centroid. Understanding how load paths work through thin-walled structures — how bending loads are carried by the spars and skin acting as a box beam, how shear loads are distributed around the closed cross-section, how rib loads are reacted into the spar webs, and how the skin panels carry compression loads up to the buckling limit — is the structural mechanics understanding that allows a structural designer to create geometry that works structurally rather than geometry that merely fits. Designers who understand why the rib pitch is what it is (the compression buckling constraint on the skin panel between ribs), why the spar flanges are as wide as they are (the fastener edge distance and pitch required for the flange load), and why the skin thickness changes along the span (the bending moment envelope and the weight optimisation) are demonstrating the structural awareness that distinguishes structural designers from CAD operators.
- **Fastener design and joint analysis — the most detail-intensive aspect of structural design.** Fastened joints — the interfaces where individual structural components are connected by rivets, bolts, or hi-lok fasteners — are the most numerically dense aspect of aerospace structural design, requiring the designer to specify the fastener type, diameter, material, head style, length, and surface finish for every fastener

in the assembly, arrange the fasteners in patterns that satisfy edge distance, pitch, and row spacing requirements, and ensure that the joint produces the load transfer required by the stress analysis. Understanding the fastener design rules that govern structural joints — minimum edge distance (typically $2D$, where D is the fastener diameter), minimum pitch (typically $3D$ between adjacent fasteners in a row), the bearing-bypass interaction in composite laminates, the fretting fatigue risk at metal-metal interfaces, and the sealing requirements for fuel-tank and pressurised-structure applications — is the joint design competency that is specifically tested in structural design interviews at every level. Designers who have specified fastener patterns for primary structural joints — and who have navigated the interaction between the stress analyst's load transfer requirement and the manufacturing team's access and installation constraints — are demonstrating the detailed design competency that aerospace roles require.

- **Composite structural design — the material that now dominates primary aircraft structure.** Carbon fibre reinforced polymer (CFRP) composites constitute the majority of primary structure on all new-generation commercial aircraft (A350, 787, A220) and an increasing proportion of defence and spacecraft structure. Designing in composite introduces constraints that metallic design does not have: the ply orientation must be specified for every ply in the laminate stackup (0° , $\pm 45^\circ$, 90°), and the resulting stiffness and strength is anisotropic — the structure is much stronger in the fibre direction than transverse to it. The minimum ply percentage rules (typically a minimum of 10% of plies in each of the four ply orientations: 0° , $+45^\circ$, -45° , 90°) ensure that the laminate has sufficient off-axis stiffness to carry multi-directional loads. The joint design rules for composite structures differ from metallic rules: bearing stress limits are lower, the bypass stress interaction is more complex, and the pull-through failure mode at fastener locations requires specific local reinforcement. Understanding how to design in composite — how to specify a laminate, how to design a composite-to-metal interface, and how to manage the consequences of impact damage and delamination on the structural design — is the material competency that is now required at every level of structural design engineering.
- **Design for manufacture and assembly — the constraint that turns a structurally efficient design into a producible component.** A structurally elegant design that cannot be manufactured to the required tolerance, assembled in the correct sequence without interference, or inspected with the specified technique is not a finished engineering design — it is a structural concept that still requires engineering. Design for manufacture (DFM) in aerospace structural design involves: understanding the capabilities of the manufacturing process (what features a CNC machining centre can and cannot produce, what draft angles a casting requires, what the minimum bend radius is for a sheet metal form), designing features that allow the manufacturing tooling to locate, clamp, and process the component consistently, and avoiding features that produce stress concentrations at locations where fatigue damage is likely. Design for assembly (DFA) involves: sequencing the structural assembly in an order that allows each component to be installed and fastened without interference from components that will be installed later, ensuring that the fastener installation tools have clearance to reach every fastener location, and designing the shimming and adjustment provisions that allow the assembly to be closed to tolerance even when individual components are at the limits of their manufacturing tolerance.
- **Configuration management and the engineering change process — the discipline through which design changes are controlled.** An aerospace structural component in production exists in a precisely defined configuration — a specific revision of the CAD model, the drawing, the material specification, and the manufacturing process specification. Any change to any of these — a geometry modification driven by a stress finding, a material substitution driven by a supply chain issue, a tolerance change driven by a manufacturing non-conformance — must be managed through a formal engineering change process that ensures the change is reviewed, authorised, documented, and reflected in the configuration baseline before it is implemented. Understanding how the engineering change process works in an aerospace environment — the difference between a Class I change (which requires authority approval because it affects airworthiness, weight, or performance) and a Class II change (which can be approved internally), what an engineering concession is and how it is managed, and how the CAD configuration management system (the Product Lifecycle Management system — Dassault's ENOVIA, Siemens' Teamcenter, PTC's Windchill) tracks and controls the released configuration — is the engineering governance competency that structural design engineers must have from their first year in a production programme.
- **Structural design for certification — how the physical design must be documented to support the**

airworthiness substantiation. The structural design of a certified aircraft component is not complete when the CAD model is geometrically correct — it is complete when every structural feature can be substantiated by the stress analysis, the material data, the manufacturing specification, and the test evidence required by the certification basis. Understanding how the structural design package supports the certification substantiation — what design features must be captured in the structural arrangement drawings that the stress analysis references, what the material specification must define about the material's strength allowables, and how the as-manufactured geometry (the inspection report and the first article inspection data) is reconciled against the nominal design geometry in the stress analysis — is the certification design literacy that structural design engineers at OEMs and major aerostructures suppliers must develop.

How Structural Design Engineering Interviews Differ from Structural Analysis Roles and from Mechanical Engineering Generally

- **The output is geometry, not numbers.** Structural analysis produces a margin of safety — a number that tells you whether the structure is strong enough. Structural design produces geometry — the physical shape that the analysis will evaluate. The structural design engineer who thinks in margins will be constantly dependent on the stress analyst to tell them whether their design works; the structural design engineer who can read a load case and sketch an efficient load path before the analysis is run is demonstrating the geometric-structural reasoning that the discipline requires. Interviews will specifically probe whether the candidate's understanding of structures drives their design intuition, or whether they design and then wait for analysis to validate.
- **The constraints are always in conflict.** Every structural design decision is made under simultaneous constraints that pull in different directions: strength requires more material, weight requires less; stiffness requires thick sections, manufacturing requires thin features that can be formed; corrosion protection requires surface treatments that affect assembly tolerance; inspection access requires open structure that reduces stiffness. The structural design engineer who cannot navigate these conflicts — who resolves them by always deferring to one constraint at the expense of the others — will produce designs that are structurally sound but too heavy, or light but unmaintainable, or analysable but unbuildable. The ability to find design solutions that satisfy multiple competing constraints simultaneously — and to recognise when a constraint is genuinely binding versus when a creative geometric solution can relax it — is the design intelligence that structural design employers specifically value.
- **The CAD model is the engineering product.** Unlike analysis, where the engineering judgement is expressed in a calculation note, structural design engineering's primary output is the CAD model — the parametric 3D geometry that defines the component to the precision required for manufacturing. The quality of that model — its geometric accuracy, its parametric robustness, its organisation and naming conventions, its compliance with the employer's modelling standards — is a direct expression of the engineer's professional discipline. Interviewers will assess not just whether the candidate can create geometry, but whether they create geometry that others can understand, modify, and build from without the originating engineer present to explain it.
- **The manufacturing and assembly teams are constant partners.** A structural design engineer who designs in isolation — who produces a CAD model without engaging the manufacturing engineer, the tooling designer, and the assembly mechanic — will produce designs that fail at the producibility review and require rework that costs time, money, and credibility. The structural design function exists at the intersection of structural analysis, manufacturing engineering, and assembly operations, and the engineer who can work effectively across these three adjacent functions — who understands enough about machining to know what features drive cost, enough about composite layup to know what ply schedules are practical, and enough about assembly operations to know which fastener locations are inaccessible — is the structural design engineer who produces designs that get built the first time.
- **Spatial reasoning is the baseline competency.** Most engineering disciplines require strong mathematical reasoning. Structural design engineering requires strong mathematical reasoning and strong spatial reasoning — the ability to visualise three-dimensional geometry from two-dimensional representations, to understand how a complex assembly moves through its degrees of freedom, and to anticipate interference conditions that will not appear in the CAD model until the assembly is mated at the wrong point

in the development process. The interview will probe spatial reasoning — sometimes explicitly through geometric problems, sometimes implicitly through questions about the candidate's most complex modelling experience and what the geometric challenges were.

PART II: 20 SPECIFIC PREP TASKS

INTERN / GRADUATE

1. Understand the structural design ecosystem — the loads, the design, the analysis, the manufacturing, and how they interconnect 4–5 hours

Skill: The system-level map of the structural design function that frames every engineering contribution within it

Why the field cares: Structural design does not exist in isolation — it is the central discipline that receives requirements from loads and aeroelastics, translates them into geometry for stress analysis, negotiates the producible form with manufacturing engineering, and provides the certified definition to the production line. The structural designer who understands only the CAD model — and not the technical ecosystem within which the model is produced and used — will create geometry that satisfies the immediate geometric requirement but misses the downstream consequences. The designer who does not understand that changing the rib flange geometry will require a new stress substantiation, a new manufacturing plan, and a new tooling design — each of which takes time that may not be available in the programme schedule — will make geometric decisions that are locally sensible but globally damaging. Understanding the full technical ecosystem — loads → design → stress → manufacturing engineering → tooling → production → inspection → certification — is the programme awareness that distinguishes structural design engineers who create value from those who create rework.

Done signal: You can describe the structural design ecosystem's seven major stakeholder functions — loads and aeroelastics, structural design, stress analysis, manufacturing engineering, tooling design, quality and inspection, and airworthiness/certification — and for each, state what it must provide to the structural design function, what it receives from the structural design function, and one specific consequence for the downstream function if the structural designer makes an error in the interface between them. You can explain why a 2mm change to a rib flange geometry — apparently trivial — could require a stress re-analysis, a tooling modification, a manufacturing plan update, and a configuration baseline change, each with its own lead time and cost.

2. Develop working proficiency in your primary CAD platform — CATIA V5, CATIA V6/3DEXperience, Siemens NX, or SolidWorks — at the level of structured solid and surface modelling 10–20 hours

Skill: The primary technical tool of the structural design discipline

Why the field cares: CAD proficiency in the aerospace-standard platform is the baseline competency without which no structural design role can be performed. CATIA V5 remains the dominant platform at Airbus, Spirit AeroSystems, GKN Aerospace, and the majority of the European and US aerostructures supply chain. CATIA V6 / the 3DEXperience platform is being progressively adopted across Airbus programmes and at some Boeing suppliers. Siemens NX is the primary platform at several defence primes and at Lockheed Martin. SolidWorks is used at smaller manufacturers, advanced air mobility developers, and in academic and early-career contexts. Understanding the platform at the level of its structural design workflows — not just its basic modelling operations — means understanding how to create parametric geometry that updates correctly when a driving dimension changes, how to manage a multi-component assembly with correct constraint structure, how to apply GD&T annotations in a model-based definition environment, and how to produce engineering drawings from the 3D model that meet the employer's drawing standard. Employers assess this by looking at the portfolio of models the candidate has created, the complexity of the assemblies they

10–20 hours

have managed, and the questions they ask about the employer's CAD environment.

Done signal: You have a portfolio of at least three CAD models of increasing complexity that you have created in your primary platform: a simple machined component with holes, flanges, and pockets; a sheet metal or composite panel with formed features; and an assembly of at least five components with correctly defined constraints, clearances, and at least one moving joint. You can describe the modelling strategy you used for each — the feature tree structure, the parametric relationships, the reference geometry — and explain one specific geometric challenge each model presented and how you resolved it. You can describe how your primary CAD platform handles model-based definition and what its approach to configuration management is.

4–5 hours

3. Understand thin-walled aerospace structural elements — spars, ribs, frames, stringers, and skin panels — at the level of their geometry, function, and load-carrying role

Skill: The structural vocabulary that allows a structural designer to create geometry with structural intent

Why the field cares: A structural designer who models a wing rib as a solid block of material — because that is what satisfies the geometric envelope constraint — without understanding why ribs are webbed structures with lightening holes, flanges, and cruciform stiffeners, is producing geometry without structural intent. The geometry of each structural element in an airframe is driven by its structural function: the spar cap carries the bending moment as an axial force (high area, concentrated at the outer surface of the cross-section); the spar web carries the shear (thin sheet, stiffened against buckling by vertical stiffeners); the rib web carries the local rib load and reacts the skin panel buckling load (thin sheet, lightened by holes that reduce weight without significantly reducing shear stiffness because the shear is carried in the periphery); the stringer carries compression and tension in combination with the skin panel (T or Z section, designed to delay the skin panel buckling by reducing the panel width). Understanding why each element has the geometry it has — and what would happen structurally if the geometry were changed — is the structural awareness that transforms a CAD operator into a structural design engineer.

Done signal: You can describe the structural function and typical geometry of each of the seven primary airframe structural elements (spar caps, spar webs, ribs, frames, stringers, skin panels, and primary attachment fittings) and explain why the geometry of each is driven by its structural function. You can explain why a rib web has lightening holes, what constraints determine the maximum hole diameter relative to the web height, and why the holes are typically located near the neutral axis of the rib rather than near the flanges. You can sketch a typical wing box cross-section, label its structural elements, and describe the load path for a downward bending load applied at the wing tip.

2–3 hours

4. Prepare behavioural answers demonstrating spatial problem-solving, manufacturing awareness, and design iteration under constraint

Skill: The character signals that structural design employers specifically probe

Why the field cares: Structural design employers assess whether candidates think in three dimensions naturally, whether they engage constructively with manufacturing and assembly constraints rather than treating them as obstacles, and whether they iterate their designs efficiently in response to analysis feedback or geometric conflicts. The stories that are weakest in structural design interviews describe CAD modelling in isolation ("I modelled a wing rib in CATIA"). The strongest describe design problem-solving under constraint ("I found that the rib flange I had designed for the lower skin interface would interfere with the hydraulic line routing in three of the seven rib bays, and I worked with the systems team to reroute two of the lines and redesigned the rib flange as a locally jogged feature in the third bay where rerouting was not possible, then updated the stress analysis to check the joggle's effect on the flange axial load capacity"). The candidate who has engaged with real structural design challenges — who can describe a specific geometric conflict, the constraints it imposed, and the design solution they found — is demonstrating the spatial engineering intelligence

2–3 hours

that structural design employers look for.

Done signal: Four STAR stories covering: a situation where you identified a geometric conflict or interference in a structural assembly and resolved it through design iteration; a time you found that your initial design was not manufacturable and redesigned it to meet the manufacturing constraint without significantly compromising structural performance; a situation where you worked with an adjacent engineering discipline (stress, manufacturing, systems, weight) to resolve a conflict between their requirement and your geometric design; and a time you had to modify a design after it was partially through the release process and managed the change through the configuration management system. At least two stories involve specific geometric content — a specific feature, a specific conflict, a specific design decision — not generic process description.

4–5 hours

5. Understand GD&T and tolerance stack-up — the language that connects the design to manufacturing and inspection

Skill: The geometric communication competency without which a structural design package is incomplete

Why the field cares: A CAD model without correct GD&T is an incomplete engineering definition — it describes the nominal geometry but does not communicate how much variation the design can tolerate. In an assembly where multiple components must fit together to achieve a specific interface gap, the manufacturing tolerances on each individual component accumulate into a worst-case assembly gap that may be larger or smaller than the design requires. Understanding how to perform a tolerance stackup analysis — identifying the contributing dimensions, applying their tolerances in the worst-case or statistical sense, and assessing whether the resulting assembly gap meets the design requirement — is the design mathematics that prevents interference conditions and inspection failures from appearing in production. GD&T in aerospace structural design also governs how the tooling is designed to locate and constrain the component, how the inspection equipment measures compliance, and how a non-conformance (a dimension measured outside its tolerance) is dispositioned against the structural requirement.

Done signal: You can explain the difference between a dimensional tolerance (a $\pm$ value on a linear dimension) and a geometric tolerance (a GD&T callout that controls shape, orientation, or position) and describe one scenario in which a dimensional tolerance alone is insufficient to define the design intent. You can explain what a true position callout for a fastener hole pattern means — what the tolerance zone is, what datum reference frame it is measured from, and how it is verified during inspection. You can perform a simple one-dimensional worst-case tolerance stackup for a three-component assembly — identifying the contributing dimensions, applying the worst-case tolerances, and determining the resulting gap or interference at the interface.

5–6 hours

6. Get hands-on with aerospace structural design references — CS-25/FAR Part 25, MMPDS, and the structural design handbooks relevant to your platform

Skill: Specific technical engagement with the field's governing standards and material data

Why the field cares: Unlike many engineering disciplines where the practitioner's handbook knowledge is largely absorbed through coursework, aerospace structural design has a specific set of governing documents that every professional must engage with directly. CS-25 (EASA) and FAR Part 25 (FAA) are the airworthiness standards that define the load cases, structural performance requirements, and certification methods that the structural design must satisfy — the designer who does not know what CS-25.301 requires (limit loads and ultimate loads) or what CS-25.571 requires (damage tolerance and fatigue evaluation) cannot design with the correct structural intent. MMPDS (Metallic Materials Properties Development and Standardization, formerly MIL-HDBK-5) is the FAA-accepted source of structural material allowables for metallic aerospace materials — the designer who selects a material without checking its MMPDS-basis allowables (A-basis or B-basis, depending on the criticality of the application) is designing without the correct material properties. The composite design handbook CMH-17 (Composite Materials Handbook, formerly MIL-HDBK-17) provides the equiva-

5–6 hours

lent material and design guidance for composite structures.

Done signal: You have read CS-25 Subpart C (Structure), specifically CS-25.301 (Loads), CS-25.303 (Factor of Safety), CS-25.305 (Strength and Deformation), CS-25.571 (Damage Tolerance and Fatigue Evaluation), and CS-25.613 (Material Strength Properties and Design Values), and can describe without consulting the document: what the limit load and ultimate load definitions are and what structural behaviour each requires, what the standard factor of safety is and when it may be modified, what the damage tolerance requirement means for the design of a primary structure inspection programme, and how A-basis and B-basis material allowables are defined and when each is required. You have accessed MMPDS and looked up the allowables for at least one common aerospace aluminium alloy (2024-T3 or 7075-T6) and can describe what the F_{tu} , F_{ty} , F_{cy} , F_{su} values represent.

MID-LEVEL (~3–8 years)

All graduate tasks still apply. Design stories must come from professional programme contexts. Technical knowledge must extend to specific design decisions, their structural rationale, and their manufacturing and certification consequences.

5–6 hours

7. Build thorough working knowledge of fastened joint design — edge distances, pitch, bearing-bypass interaction, and the specific rules for composite joints

Skill: The most detail-intensive and most frequently tested aspect of structural design competency

Why the field cares: Fastened joints are the primary interface between structural components, and they are simultaneously the most common source of structural design errors and the most heavily scrutinised aspect of the stress substantiation. Every fastener in a primary structural joint must satisfy: minimum edge distance (the distance from the fastener centre to the nearest free edge, typically 2.0D minimum to prevent bearing failure at the edge), minimum pitch (the distance between adjacent fasteners in a row, typically 3.0D to prevent interaction between adjacent bearing failure zones), maximum pitch (the distance between rows, governed by skin panel stability between fastener rows), and the correct bearing area based on fastener diameter and material thickness. In composite joints, the bearing-bypass interaction — the combined effect of the stress concentration at the fastener hole and the bypass load that must be carried around the joint — produces a more complex failure criterion than metallic joints, and the designer must ensure that the laminate has sufficient fibre content in the load direction to carry both. Designers who have specified fastener patterns for primary structural joints — who have worked through the edge distance and pitch calculations, resolved the access constraints with the assembly team, and seen the stress analyst's bearing-bypass check — are demonstrating the joint design depth that mid-level structural design roles require.

Done signal: You can explain the minimum edge distance and pitch requirements for a metallic structural joint and describe what failure mode each requirement is designed to prevent. You can explain the bearing-bypass interaction in a composite joint — what the two load components are, how they interact in the failure criterion, and why the composite laminate's ply orientation affects the allowable bearing stress. You can describe how to select the correct fastener diameter for a given structural application — what the governing constraint is (bearing area, shear strength of the fastener, or pull-through of the composite laminate) and how the diameter selection is confirmed with the stress analyst.

5–6 hours

8. Understand composite structural design in depth — laminate design, ply orientation rules, co-cured versus fastened joints, and impact damage considerations

Skill: The material competency that is now required for primary structure design on all new-generation platforms

Why the field cares: Carbon fibre composite is the dominant structural material for primary structure on new-generation commercial aircraft and a rapidly growing proportion of defence and spacecraft structure. Designing in composite introduces constraints that metallic design does not have: the ply

5–6 hours

orientation must be specified for every ply in the laminate stackup (0° , $\pm 45^\circ$, 90°), and the resulting stiffness and strength is anisotropic — the structure is much stronger in the fibre direction than transverse to it. The minimum ply percentage rules (typically a minimum of 10% of plies in each of the four ply orientations: 0° , $+45^\circ$, -45° , 90°) ensure that the laminate has sufficient off-axis stiffness to carry multi-directional loads. The joint between a composite component and an adjacent metallic or composite component is the most structurally critical and most design-intensive aspect of composite structural design — whether co-cured (the joint is created during the cure process, without fasteners), co-bonded (one component is bonded to a cured component during its own cure cycle), or secondarily bonded or fastened (the two cured components are joined by adhesive or fasteners). Understanding the design rules for each joint type, the inspection methods that validate each, and the certification implications of each are the composite design competencies that mid-level structural designers on new-generation programmes must have.

Done signal: You can describe the three ply orientation rules for a balanced symmetric composite laminate — the minimum percentage rule, the symmetry rule, and the balance rule — and explain why each is required. You can explain why a quasi-isotropic laminate (25% in each of 0° , $+45^\circ$, -45° , 90°) is often used as a starting point for composite structural design even though it is not weight-optimised, and what the structural trade-off is between a quasi-isotropic laminate and a laminate optimised for a specific load direction. You can describe the structural difference between a co-cured joint and a secondarily fastened composite-to-composite joint, and explain one specific failure mode in each type of joint that the structural design must be configured to prevent.

2 hours

9. Prepare a technically specific story about a structural design conflict you resolved — where geometry, structure, and another engineering domain were in direct conflict

Skill: The design problem-solving under multi-constraint pressure that distinguishes effective structural design engineers

Why the field cares: The defining challenge of structural design engineering is not creating geometry — it is creating geometry that satisfies simultaneous constraints from multiple engineering domains. Every structural design engineer who has worked on a real programme has encountered the situation where the stress analyst needs more material, the weight engineer says there is none available, the manufacturing engineer says the current geometry is not producible with the selected process, and the systems team says the space that would allow the structural solution is already occupied by a hydraulic line. The resolution of this kind of multi-domain conflict — through a specific geometric insight, a material grade change, a manufacturing process modification, or a systems reroute — is the design problem-solving that structural design employers value most and that is hardest to demonstrate without a specific story.

Done signal: You have a story about a structural design conflict you resolved — in a professional programme context — where the geometric requirement was in direct conflict with at least two other engineering domain requirements (weight, stress, manufacturing, systems, inspection, or certification). The story describes the specific conflict (the components involved, the geometric constraint, the competing requirements), the design options you explored (what geometric approaches you considered and why each did, or did not, resolve the conflict), the solution you implemented (the specific geometric feature or design change), and the validation step (what analysis or test confirmed that the solution satisfied all of the competing constraints). The story names the aircraft type, the structural element, and the specific geometric feature — it is not generic.

4–5 hours

10. Understand the structural design for damage tolerance — crack growth, inspection intervals, and what the design must provide to support the damage tolerance substantiation

Skill: The certification-driven structural design requirement that governs primary structure design on civil transport aircraft

Why the field cares: CS-25.571 requires that primary structure on civil transport aircraft be designed to damage tolerance — the structure must be able to sustain damage (a fatigue crack, a corrosion pit,

4–5 hours

or an impact dent) of a specified size and continue to carry limit load until the damage is detected by the scheduled inspection programme. This requirement has specific implications for the structural design: the structural arrangement must provide crack arrest features (tear straps, spanwise joints, or multi-load-path designs) that prevent a fatigue crack from growing to critical size between inspection intervals, the inspection technique must be capable of detecting the critical crack size at the specified inspection threshold, and the structural design must allow access for the specified inspection technique. Understanding how the damage tolerance requirement drives structural design decisions — why a skin panel has a maximum bay length (the crack arrest strap pitch), why a fastener row at a structural joint is spaced to provide a crack initiation site rather than allowing a continuous crack path, and why the structural arrangement must be documented with sufficient detail to support the damage tolerance analysis — is the certification design awareness that primary structure designers on civil programmes must develop.

Done signal: You can explain what CS-25.571 requires for the damage tolerance evaluation of primary structure — what “principal structural elements” are, what the two-load-path requirement means, and what the inspection programme must demonstrate about the damage detection capability. You can describe what a crack arrest strap is, where it is placed in a wing skin panel design, and what structural design parameter determines its pitch. You can explain how the choice between a safe-life design approach and a damage tolerance approach affects the structural design — what the safe-life approach requires in terms of fatigue testing, what the damage tolerance approach requires in terms of structural arrangement and inspection programme design, and why civil transport aircraft are required to use damage tolerance rather than safe-life for primary structure.

4–5 hours

11. Understand the engineering change process in an aerospace production programme — Class I versus Class II changes, engineering concessions, and the CAD configuration management workflow

Skill: The engineering governance competency that determines whether design changes are managed safely and efficiently

Why the field cares: In a production aerospace programme, a structural design change is not complete when the CAD model is updated — it is complete when the change has been reviewed, classified, authorised, documented in the configuration baseline, communicated to all affected downstream functions, and implemented on the production line without introducing a non-conformance. Understanding the engineering change process — how a change is initiated (by a stress finding, a manufacturing non-conformance, a supplier deviation, or a service bulletin), how it is classified (Class I if it affects airworthiness, weight, performance, or interchangeability; Class II if it does not), what the review and authorisation process is for each class, and how the CAD configuration management system tracks the change from initiation to closure — is the engineering programme management competency that structural design engineers must have by the mid-level. The specific change management tools used in aerospace programmes — ENOVIA for Airbus programmes, Teamcenter for many Boeing and defence programmes, Windchill for others — vary by employer, but the underlying process discipline is consistent.

Done signal: You can describe the engineering change process in an aerospace production programme — from change initiation through classification, review, authorisation, documentation, and implementation — and identify the key decision points and the stakeholders involved at each. You can explain the difference between a Class I and a Class II change in terms of the approval authority required and the downstream notification obligations. You can describe what an engineering concession is — a temporary authorisation to use, repair, or release a non-conforming component — and explain how a concession differs from a change in terms of its permanence and its configuration management treatment. You can describe how the PLM system you have used tracks the relationship between a change and the affected CAD model revisions, drawings, and manufacturing plans.

4–5 hours

12. Build working knowledge of structural design for spacecraft and launch vehicles — where the constraint environment differs from aircraft

Skill: The domain extension that allows structural design engineers to work across both aviation and space sectors

Why the field cares: Spacecraft and launch vehicle structural design shares its foundational principles with aircraft structural design — thin-walled structures, load path management, joint design, damage tolerance — but operates under a constraint environment that differs in several important ways. The load cases are different: launch vehicles experience high axial compression during powered flight, acoustic loads during the first 90 seconds of flight that are more severe than anything a transport aircraft encounters, and thermally induced loads from cryogenic propellant tanks adjacent to aluminium structure. The material selection environment is different: mass is even more critical than in aircraft design, driving the use of carbon-carbon composites, beryllium, and high-specific-strength alloys that are rarely used in civil aviation. The certification approach is different: spacecraft are not certificated under CS-25 or FAR Part 25, but under a combination of NASA standards (NASA-STD-5001 for metallic structures, NASA-STD-5020 for threaded fastening systems), ESA standards (ECSS-E-ST-32 for structural engineering), and programme-specific design requirements. The design life is different: a spacecraft may be designed for a single mission (an expendable launch vehicle upper stage), a defined number of missions (a reusable crew vehicle), or an extended on-orbit life (a communications satellite platform). Understanding these distinctions — and how they change the structural design approach — is the domain extension competency that allows structural design engineers to move effectively between aviation and space programmes.

Done signal: You can describe three specific ways in which the structural design constraint environment for a spacecraft or launch vehicle differs from that for a civil transport aircraft — identifying a specific load case, a specific material consideration, or a specific certification approach that is unique to the space domain. You can explain what acoustic fatigue is, why it is a critical design case for launch vehicle payload fairings, and what structural design features are used to manage it. You can describe what the NASA-STD-5001 metallic structural design standard requires in terms of design factor of safety and its rationale, and how this compares to the CS-25 factor of safety for civil transport aircraft primary structure.

4–6 hours

13. Prepare for a cross-domain structural design problem under interview conditions — a geometric conflict scenario or a design-change impact analysis with cascading engineering consequences

Skill: Structured structural design problem-solving under time pressure

Why the field cares: Mid-level and senior structural design engineering interviews consistently include a cross-domain design problem — a geometric conflict or a design change that requires reasoning across structural, manufacturing, weight, and systems engineering domains simultaneously. “Your forward spar web has been increased in thickness by 1.5mm at the root bay by the stress team following a revised gust load case — walk through the impacts on the rib foot fitting geometry, the spar-to-fuselage attachment fitting interface, the fastener pattern at the web-to-cap interface, the weight budget, and the manufacturing plan for the current tooling.” The candidate who can trace these cascading implications — thickness increase → rib foot fitting depth → spar frame interface position → fastener pattern spacing → weight delta → tooling interference check — is demonstrating the integrated design thinking that structural design engineering requires.

Done signal: You can work through a structural design impact analysis problem in under 20 minutes — tracing the engineering consequences of a design change (a dimension change, a material substitution, a load case revision, or an interface geometry modification) through at least three engineering domains (structural geometry, fastener design, weight, manufacturing, systems interface, or certification). You have practised this for at least two different scenarios. You articulate each engineering consequence explicitly, identify the key constraint at each stage, and describe what information you would need from adjacent engineering functions to complete the impact assessment.

SENIOR LEVEL (Principal Engineer, Lead Designer, Chief of Structure)

All mid-level tasks still apply. Technical knowledge must extend to structural architecture decisions at the assembly level. Design leadership experience must include managing the geometry across a team of designers and representing the structural design position in multi-discipline integration reviews.

14. Prepare to walk through the most complex structural assembly you have personally led — the architectural decisions, the interface management, and the certification substantiation it supported 4–6 hours

Skill: Senior structural design leadership credibility demonstrated through a specific and architecturally significant design

Why the field cares: Senior structural design engineers are specifically valued for having led the design of a major structural assembly — an assembly complex enough to involve multiple load paths, multiple material interfaces, multiple manufacturing processes, and multiple adjacent engineering domains competing for the same geometric envelope. The story must demonstrate not just that a design was produced, but that specific architectural decisions were made — the load path philosophy was chosen from alternatives and the choice was defended, the interface control document was established and managed through the programme, the design was iterated through analysis feedback and manufacturing review and emerged as a certified structural definition. Engineers who have done this, and who can describe the specific architectural decisions and why they made them, have experience that is genuinely formative and that cannot be credibly invented.

Done signal: For one major structural assembly you personally led the design of — an assembly involving at least three structural components, at least two interface types (fastened, bonded, or welded), and at least two adjacent engineering domains — you can describe: the structural architecture you chose (the load path philosophy, the primary structural elements, the joint philosophy), why you chose it over the alternatives you considered, the most difficult interface management challenge you encountered (and how it was resolved), the most significant design change that was imposed after the initial design was established and how you managed its impact across the assembly, and the certification substantiation that your design supported. You can describe one decision you would make differently with the benefit of hindsight, and why.

15. Prepare a technically grounded position on one active structural design frontier — additive manufacture for structural components, digital twin integration, or advanced composite architecture 4–5 hours

Skill: Forward-looking technical awareness at senior engineering depth

Why the field cares: Structural design engineering in 2025–2026 has several genuine technical frontiers where the design methodology, the certification approach, and the manufacturing technology are all in simultaneous development. Metal additive manufacture (laser powder bed fusion, directed energy deposition) is enabling structural geometries that are impossible with conventional machining — topology-optimised primary fittings, integrated cooling channels in thermally loaded structure, consolidated multi-part assemblies — but the certification basis for additively manufactured primary structure is still being developed, and the material allowables are not yet available in a handbook-basis form. Advanced composite architectures — automated fibre placement enabling variable angle tow laminates that optimise fibre orientation along the load path, out-of-autoclave processing enabling large-format composite structures, and thermoplastic composite enabling welded joints that eliminate the fastener pattern entirely — are transforming composite structural design in ways that the existing design handbooks do not fully address. Digital twin integration — the coupling of the structural CAD model with real-time strain and load data from the in-service aircraft to update the damage tolerance analysis — is changing the relationship between the structural design definition and the in-service structural management.

Done signal: You have a prepared 3-minute technical discussion covering one specific frontier — named specifically: the certification challenges for additively manufactured primary structure and what the current FAA/EASA special conditions require in terms of process qualification, material

4–5 hours

allowables, and non-destructive inspection; the variable angle tow composite design approach and what structural efficiency gain it demonstrates relative to conventional fixed-angle laminates in a representative primary structure application; or the digital twin integration approach and how it changes the basis for the structural inspection programme — specifically what data from the in-service aircraft changes which parameter in the damage tolerance model. Your position identifies the specific unresolved technical challenge and describes the approaches being taken to address it.

4–5 hours

16. Prepare to lead a technical discussion on a structural architecture decision — multi-spar versus two-spar wing box, metallic versus composite primary structure, or monolithic versus built-up construction

Skill: Strategic structural design decision-making at the architecture level

Why the field cares: Senior structural design engineers are called on to make or advise on structural architecture decisions that have decade-scale manufacturing and operational consequences. “For a new regional aircraft with a design range of 2,000 nm, a maximum takeoff weight of 75,000 kg, and a target entry into service in 2029, recommend whether the wing box primary structure should be metallic (aluminium-lithium alloy) or composite (CFRP), given the programme’s certification schedule, the available manufacturing technology at the primary aerostructures supplier, and the weight and maintenance cost targets.” This requires reasoning through: the structural efficiency comparison between the two material systems at the relevant scale, the certification timeline difference between a metallic wing box (established means of compliance) and a composite wing box (extensive testing programme required), the manufacturing readiness of the aerostructures supplier for each material system, and the through-life maintenance cost difference (composites resist corrosion but are more complex to repair). The senior interview tests whether you can structure this reasoning live and arrive at a specific recommendation.

Done signal: You have worked through at least two structural architecture trade-off problems — one involving a material selection decision and one involving a structural arrangement decision (load path philosophy, joint type, or construction approach). For each, you state the governing constraints, the two or three credible options, the key trade-off, and your recommendation with explicit sensitivities to the assumptions. You can do this in under 10 minutes per scenario, and you can identify which assumption is most load-bearing for your recommendation.

4–5 hours

17. Understand how to manage a structural design team through a major design cycle — the review sequence, the interface control process, and the integration with adjacent engineering functions

Skill: The engineering leadership competency that determines whether a structural design team produces a coordinated assembly

Why the field cares: A structural assembly designed by a team of five engineers will only be geometrically coordinated if the team leader has established a rigorous interface control process — a documented definition of the geometry at every interface between components, agreed by the designers on each side of the interface before the detailed design begins, and managed as a controlled baseline throughout the design cycle. Without this process, each designer optimises their own component correctly but the interfaces between components are discovered as conflicts at the integration review, when changes are expensive and schedule-impacting. The senior structural design engineer who has led a team through this process — who has defined the interface control documents, managed the interface baseline, run the design review cycle (PDR → DDR → CDR → first article inspection), and represented the structural design position in the multi-discipline integration reviews — has the engineering management experience that senior structural design roles require.

Done signal: You can describe the structural design review sequence for a major structural assembly — from the conceptual design review (CDR) through the preliminary design review (PDR) to the detailed design review (DDR) and the manufacturing readiness review (MRR) — identifying what is baselined at each review gate, what the review criteria are, and what design maturity is required before each gate is passed. You can describe how an interface control document is structured — what

4–5 hours

geometry is captured, how changes to the baselined interface are managed, and who has authority to approve an interface change. You can describe the specific integration activities at each design phase — when the structural design model is checked against the systems routing models, when the tooling design team receives the structural geometry, and when the manufacturing engineering team is engaged on the producibility review.

4–5 hours

18. Build quantitative estimation fluency for structural design trade-offs — weight estimates, margin calculations, and fastener count estimates — without a full stress analysis

Skill: Senior-speed first-principles structural reasoning that enables rapid architecture-level decisions

Why the field cares: Senior structural design engineers are expected to provide rapid structural assessments of design alternatives before the full stress analysis is performed — in a design review, in a weight review, or in an architecture trade study. “The stress team is proposing to increase the lower spar cap area by 15% to address a margin shortfall at the root section — what is your estimate of the weight impact on the spar assembly, and what alternative approaches to recovering the margin would you want to assess before accepting the weight penalty?” should produce a calibrated estimate and a structured set of alternatives within minutes, not a request for a week of CAD and analysis work. These estimation problems are tractable from first principles: the weight of a spar cap increase can be estimated from the density of the material, the area increase, and the length of the affected section; the alternative approaches (increasing the cap width versus the thickness, changing the material alloy, adding a local pad-up, redistributing load to the web through fastener pattern change) can be structured from an understanding of the structural mechanics and the manufacturing constraints.

Done signal: You have worked through five structural design estimation problems covering at least two of: estimating the weight impact of a specified geometry change (a thickness increase, a cross-section modification, or a fastener diameter change) on a specified structural element; estimating the approximate margin of safety of a simple structural element under a specified load using a simplified analysis (beam bending, column buckling, or fastener bearing); estimating the fastener count for a specified joint length and fastener pitch; estimating the manufacturing cost impact of a specified geometry change (a tolerance tightening, a process change, or a feature addition); and estimating the inspection interval change resulting from a specified crack growth rate modification. You state your assumptions explicitly, identify the most uncertain input, and give your estimate with a stated confidence range.

ALL LEVELS

1–2 hours

19. Prepare a specific answer to “Why structural design — and why this specific role and employer?” that reflects genuine technical and programmatic engagement

Skill: Technically specific motivation that distinguishes structurally committed candidates from those who find aerospace interesting generally

Why the field cares: Structural design engineering employers receive applications from candidates who are attracted by aerospace and by engineering — two appealing fields in combination. What they are looking for is engineers who understand the specific design challenges of the role and who have a specific geometric and structural engineering contribution to make. The expected answer engages with one of: the specific structural architecture challenge of the employer’s current programme and how the candidate’s modelling or design background applies; the composite design methodology challenge the employer faces in transitioning from metallic to composite primary structure and what the candidate’s layup design or co-cured joint experience enables; the damage tolerance design challenge on a new programme and how the candidate’s understanding of crack arrest features and inspection programme design is relevant; or the additive manufacture integration challenge and what the candidate’s topology optimisation or AM process qualification experience contributes. The

1–2 hours

answer must name one specific structural design challenge, explain what makes it geometrically and structurally hard, and connect the candidate's specific background to contributing to its solution.

Done signal: You have a 90-second answer that names the specific structural design challenge most relevant to your background, explains what makes it hard (the specific geometric conflict, the material design constraint, the certification methodology gap), and connects your engineering background to contributing to its resolution. The answer demonstrates that you have researched the specific employer's current programme — its structural material philosophy, the primary structure elements currently under design, and the manufacturing process being used — and can describe one specific structural design challenge that your background is positioned to help address. The answer cannot be given unchanged to a candidate applying to any other structural design employer.

1–2 hours

20. Map your technical background to a specific employer type, programme phase, and structural domain — and understand where each employer sits in its structural design cycle

Skill: Precise self-positioning in a sector where the employer's programme phase determines the design work and the candidate characteristics required

Why the field cares: The structural design engineering sector in 2025 spans employers at completely different phases of the design cycle, and the work — and the candidate characteristics required — differs dramatically between them. An OEM in detailed design phase (Airbus on the A320 family derivative development, Boeing on the 737 MAX production support, Embraer on the E2 entry into service support) needs structural design engineers who can manage production design changes, disposition non-conformances, and support the service bulletin process — work that requires deep programme knowledge of the existing design and the ability to manage change within a tightly controlled configuration baseline. An OEM or advanced air mobility developer in preliminary or conceptual design (a new narrowbody clean sheet, a new eVTOL configuration) needs structural design engineers who can explore the design space, develop structural architecture alternatives, and build the design that will be detailed by a larger team — work that requires strong architectural thinking, rapid modelling capability, and comfort with design uncertainty. An aerostructures supplier winning a new risk-sharing package needs structural design engineers who can develop the structural definition from the OEM's aerodynamic and interface requirements — work that requires strong ownership of the design-to-specification process. A candidate who has not understood where each employer is in its design cycle — and what the specific structural design work at that phase requires — is not ready for the interview.

Done signal: You can state in under 60 seconds: the structural design employer you are targeting and its current programme phase (conceptual, preliminary design, detailed design, production support, or in-service modification), the specific structural assembly or component domain most relevant to your target role (wing structure, fuselage structure, empennage, primary attachments, landing gear structure, doors and windows, or nacelle structure), why your modelling platform experience and structural design background matches the employer's current programme phase and domain, and one specific structural design challenge the employer is currently facing — derived from publicly available information (programme status reports, conference papers, aviation press) — that your background is positioned to help address.

SUMMARY REFERENCE TABLE

#	Task Description	Target Level	Effort
1	Structural design ecosystem — loads, design, stress, manufacturing, inspection, certification interactions	All	4–5 h
2	CAD platform proficiency — CATIA V5/V6, NX, or SolidWorks at structural design workflow level	All	10–20 h
3	Thin-walled structural elements — spars, ribs, frames, stringers, skin panels: geometry and function	All	4–5 h
4	STAR stories — geometric conflict, manufacturability, multi-domain constraint, change management	All	2–3 h
5	GD&T and tolerance stack-up — geometric tolerances, datum frames, assembly gap analysis	All	4–5 h
6	CS-25/FAR 25 and MMPDS — limit loads, ultimate loads, damage tolerance, material allowables	All	5–6 h
7	Fastened joint design — edge distance, pitch, bearing-bypass, composite joint rules	Mid+	5–6 h
8	Composite structural design — laminate design, ply orientation rules, joint types, impact damage	Mid+	5–6 h
9	Multi-domain structural conflict story — geometry, stress, manufacturing, systems in direct conflict	Mid+	2 h
10	Design for damage tolerance — crack arrest, inspection intervals, structural arrangement requirements	Mid+	4–5 h
11	Engineering change process — Class I/II changes, concessions, PLM configuration management	Mid+	4–5 h
12	Spacecraft structural design — launch loads, acoustic fatigue, NASA/ESA standards, design life	Mid+	4–5 h
13	Cross-domain structural design problem — geometry change cascade through structure, mfg, weight	Mid+	4–6 h
14	Major structural assembly leadership story — architecture decisions, interface management, certification	Senior	4–6 h
15	One active structural design frontier — additive manufacture, variable angle tow, digital twin	Senior	4–5 h
16	Structural architecture decision — material selection, load path philosophy, construction approach	Senior	4–5 h
17	Structural design team management — interface control, review sequence, multi-discipline integration	Senior	4–5 h
18	Quantitative structural estimation — weight impact, margin estimate, fastener count, inspection interval	Senior	4–5 h
19	“Why structural design — why this employer and role?” — structurally and programmatically specific	All	1–2 h
20	Employer type and programme phase positioning — OEM, aerostructures supplier, design cycle phase	All	1–2 h

THE STRUCTURAL REALITY THAT MAKES STRUCTURAL DESIGN DIFFERENT

Aerospace structural design engineering is the only discipline in which the engineer's primary output — the three-dimensional geometry of a physical structure — must simultaneously satisfy requirements from every other engineering discipline on the programme, with no priority ordering that always resolves conflicts in the same direction. The stress analyst's requirement for more material area conflicts with the weight engineer's requirement for less; the systems engineer's requirement for routing space conflicts with the structural designer's requirement for a continuous load path; the manufacturing engineer's requirement for producible geometry conflicts with the structural designer's requirement for structural efficiency; the inspection engineer's requirement for access conflicts with the structural designer's requirement for geometric closure. Every day of a structural design engineer's working life is spent navigating these conflicts — not resolving them permanently, because the constraints change as the programme evolves, but managing them to a configuration that is good enough for the current programme phase and close enough to the final answer that the downstream cost of change is manageable.

This constant conflict management — across multiple engineering domains, with a geometry that must remain coherent across thousands of interdependent features and hundreds of interface constraints — is the structural feature that distinguishes structural design from analysis. The structural analyst works on a defined geometry and computes whether it is adequate. The structural designer works on an undefined or partially defined geometry and creates it, subject to constraints that are themselves partially undefined and continuously evolving. The analyst's output is a number; the designer's output is a shape. The analyst's error is a wrong number that can be corrected by running the analysis again; the designer's error is a wrong shape that must be rebuilt from scratch, re-analysed, and re-released through the configuration management system.

The engineers who succeed in this environment are those who can hold the full geometric complexity of a structural assembly in their spatial memory — who can walk through the wing box in their mind's eye and identify where the fastener pattern is approaching minimum edge distance, where the rib foot fitting geometry will interfere with the fuel system pipework, and where the skin panel thickness change will create a step that the external aerodynamic surface cannot tolerate.

They are also those who are genuinely curious about why structural solutions take the form they do — who want to understand why the 787's wing box uses CFRP instead of aluminium-lithium, why the A350's rear spar takes the load through a cruciform fitting rather than a direct frame attachment, and what the manufacturing capability that made those solutions possible means for the design choices available on the next generation of aircraft. This geometric curiosity — about the shape, the material, the joint, and the load path — is the specific intellectual character that every serious structural design employer is trying to find in the candidates they hire, and it is what this document has been structured to help you demonstrate.

BREAK INTO AEROSPACE
